

Both experts share the same prior-selected subspace; only the structured penalty differs (ratio=0 $\equiv$ Ridge).

Ridge expert

$$\lambda_R \|\beta\|_2^2$$

$$\lambda_R \in \{10^{-3}, \dots, 10^3\}$$

NCR expert

$$\lambda_R \|\beta\|_2^2 + \lambda_L \beta^\top L_E \beta$$

$$\lambda_L = \text{ratio} \cdot \lambda_R$$



selected-edge subgraph E (prior mask)

$$\beta^\top L_E \beta = \sum_{(e, e') \in E} (\beta_e - \beta_{e'})^2 \text{ over line-graph neighbours}$$